

DTD Components (Building Blocks)

A DTD consists of several key components that define the structure and rules of the XML document:

1. Element Declaration

Declares what content an element can hold.

```
<!ELEMENT element-name content-type >
```

Content types:

- #PCDATA : Parsed character data (text)
- EMPTY : No content
- ANY : Any content
- Child elements: nested structure (e.g., (title, author))

Example:

```
<!ELEMENT book (title, author)>
<!ELEMENT title (#PCDATA)>
<!ELEMENT author (#PCDATA)>
```

DTD Element Operators

In DTD, when declaring an element with **child elements**, operators define the **order**, **optionality**, and **repetition** of those child elements.

These are also called **content model operators** and are part of the **element declaration syntax**:

```
<!ELEMENT element-name (child-elements-with-operators)>
```

List of Element Operators:

Operator	Name	Meaning	Example	Description		
,	Sequence	All listed elements must appear in order	(title, author)	<book><title>...</title> <author>...</author></book>		
	Choice	Choice (OR)	Only one of the listed elements may appear	(student teacher)		Either <student> or <teacher>
*	Zero or more	Element may appear 0 or more times	(item*)	Can have no <item> or many		
+	One or more	Element must appear at least once	(item+)	Must have at least one <item>		
?	Optional	Element may appear 0 or 1 time	(discount?)	<discount> is optional		

Examples:

1. Sequence Operator ,

```
<!ELEMENT person (name, age)>
```

Valid:

```
` <person>    <name>John</name>    <age>30</age> </person> `
```

❌ Invalid if `<age>` comes before `<name>` .

2. Choice Operator |

```
` <!ELEMENT payment (cash | card)> `
```

❌ Valid:

```
` <payment><card></card></payment> `
```

❌ Invalid if both `<cash>` and `<card>` appear.

3. Asterisk * (zero or more)

```
` <!ELEMENT itemlist (item*)> `
```

❌ Valid:

```
` <itemlist></itemlist> <itemlist><item>Apple</item><item>Orange</item></itemlist> `
```

4. Plus + (one or more)

```
` <!ELEMENT chapters (chapter+)> `
```

❌ Valid:

```
` <chapters><chapter>Intro</chapter></chapters> `
```

❌ Invalid if no `<chapter>` exists.

5. Question Mark ? (optional)

```
` <!ELEMENT user (username, email?)> `
```

❌ Valid:

```
` <user>
<username>john</username>
</user>
<user>
<username>john</username><email>john@mail.com</email>
</user> `
```

❌ Combination of Operators:

You can combine them for more complex structures:

```
` <!ELEMENT order (customer, item+, discount?)> `
```

This means:

- `customer` must appear once

- item must appear one or more times
 - discount is optional
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